

FINAL EXPLANATION: PHYSICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: PHYSICS GRADE 10: 3.2 CURRENT ELECTRICITY

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This Final Explanation brings together everything you have learned in Lessons 1 to 12. Answer each question in the space under it, using your own words, the evidence and measurements from our bench demos (9 V pack, three identical 9 V, 2.7 W, 30 Ω bulbs) and the Nairobi classroom. Show every calculation step with units. Use the terms 'in series', 'in parallel', 'potential difference (p.d.)' and 'e.m.f.' correctly. Remember that an ammeter is connected in series and a voltmeter in parallel. Your final answer should connect back to the driving question: Why do the other tubes stay lit when one blows out, and why do bulbs connected in series glow dimmer than identical bulbs connected in parallel to the same battery?

Part 1: Why the other tubes stay lit

In the Nairobi evening prep class, four fluorescent tubes share the 240 V mains and one tube blows. (a) How are the tubes connected, and what happens to the other three? (b) Using what you learned about closed loops, p.d. and current, explain why the other three tubes are not affected. (c) Each tube is 36 W on 240 V. Calculate the current in one tube, the current for all four tubes, and the current from the supply after one tube blows. (d) Compare this with an old Christmas-light string in which one bulb fails: what happens, and why?

(a) The four tubes are connected in parallel: each tube is on its own branch directly across the 240 V supply. When one blows, the other three stay lit. (b) Each tube has its own complete closed loop with the source. A blown tube breaks only its own branch, so the other branches stay closed. Each remaining tube still has the same p.d. (240 V) across it, so it still carries the same current and gives the same light. (c) $I = P/V = 36 \text{ W} / 240 \text{ V} = 0.15 \text{ A}$ for one tube. Four tubes: $4 \times 0.15 \text{ A} = 0.60 \text{ A}$. After one blows, three tubes remain: $3 \times 0.15 \text{ A} = 0.45 \text{ A}$, and each tube still carries 0.15 A. (d) The old Christmas-light string is wired in series: all the bulbs share one single loop. A failed bulb opens the only path, so the current is 0 A and the whole string goes dark.

Part 2: Why the series bulbs are dim

Mr. Otieno connected three identical 9 V, 2.7 W, 30 Ω bulbs

(a) In series, $R = R_1 + R_2 + R_3 = 30 + 30 + 30 = 90 \Omega$. (b) $I = V/R = 9 \text{ V} / 90 \Omega = 0.10 \text{ A}$. The current is the same at every point in a series loop (0.10 A), because bulbs do not use up current. (c) p.d. across each bulb = $I \times R = 0.10 \text{ A} \times 30 \Omega = 3.0 \text{ V}$. The three

in series to the 9 V battery pack. Wanjiru asked why they glow dim. (a) Calculate the total resistance of the circuit. (b) Calculate the current in the circuit and say what an ammeter would read at different points in the loop. (c) Calculate the p.d. across each bulb and show how the p.d.s relate to the e.m.f. of the pack. (d) Calculate the power in each bulb and use it to explain why they are dim. (e) What happens if one bulb is removed, and why?	p.d.s add up to the e.m.f.: $3.0 + 3.0 + 3.0 = 9.0$ V, so the energy per coulomb supplied by the pack is shared between the bulbs. (d) $P = IV = 3.0 \text{ V} \times 0.10 \text{ A} = 0.30 \text{ W}$ per bulb. This is much less than the 2.7 W rating at full 9 V, so each bulb is dim. (e) If one bulb is removed, the only path is broken (open circuit), $I = 0 \text{ A}$ and all the bulbs go out.
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Part 3: Why the parallel bulbs are bright

Now the three identical bulbs are connected in parallel to the same 9 V pack. (a) What p.d. is across each bulb, and what current flows in each branch? (b) What current comes from the battery, and what is the effective resistance of the circuit? Show it using $1/R = 1/R_1 + 1/R_2 + 1/R_3$. (c) Calculate the power in each bulb and compare it with a series bulb. (d) One bulb is removed. What are the p.d. and current in each remaining bulb, and what is the battery current now? (e) Kipchirchir predicted that the battery is used up faster in parallel. Is he right? Use your numbers.	(a) Each bulb has the full 9.0 V across it. Current in each branch $I = V/R = 9 / 30 = 0.30 \text{ A}$. (b) The battery current is the sum of the branch currents: $0.30 + 0.30 + 0.30 = 0.90 \text{ A}$. Effective resistance: $1/R = 1/30 + 1/30 + 1/30 = 3/30$, so $R = 10 \Omega$ (check: $9 \text{ V} / 0.90 \text{ A} = 10 \Omega$). (c) $P = IV = 9.0 \text{ V} \times 0.30 \text{ A} = 2.7 \text{ W}$ per bulb, full brightness. This is 9 times the 0.30 W of a series bulb ($2.7 / 0.30 = 9$). (d) The two remaining bulbs still have 9.0 V and 0.30 A each, so they stay bright. The battery current falls to $0.30 + 0.30 = 0.60 \text{ A}$. (e) Kipchirchir is partly right. The parallel circuit draws 0.90 A from the battery compared with 0.10 A in series, and its total power ($3 \times 2.7 = 8.1 \text{ W}$ against $3 \times 0.30 = 0.90 \text{ W}$) is greater, so the energy is used faster. It is only partly right because the bulbs are also much brighter.
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Part 4: Energy, cost and safe design

The school wants to light the Nairobi classroom with the four 36 W tubes in parallel on 240 V	(a) Power = $4 \times 36 \text{ W} = 144 \text{ W} = 0.144 \text{ kW}$. $E = 0.144 \text{ kW} \times 5 \text{ h} = 0.72 \text{ kWh}$. Cost = $0.72 \times \text{KSh } 25 = \text{KSh } 18$. (b) 60 W bulb: $0.060 \text{ kW} \times 5 \text{ h} = 0.300 \text{ kWh}$, cost $0.300 \times 25 = \text{KSh } 7.50$. 9 W LED: $0.009 \text{ kW} \times 5 \text{ h} = 0.045 \text{ kWh}$, cost $0.045 \times 25 = \text{KSh } 1.125$, about KSh 1.13. The LED uses far less energy and costs much less for the same time. (c) The lights should be wired in parallel across
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mains. (a) Calculate the energy used by the four tubes in 5 hours in kWh and the cost at KSh 25 per kWh. (b) A 60 W filament bulb and a 9 W LED are each used for 5 hours. Compare the energy used and the cost of each. (c) Achieng's family in Kisumu has a 12 V solar home system. Say how its lights should be connected so that one failed bulb does not put out the others, and why. (d) A kettle (3000 W) and a heater (2000 W) are plugged into one 240 V socket with a 13 A fuse. Calculate what happens and explain why. (e) Describe the safety devices that your classroom circuit should have.	the 12 V battery, so each has its own branch with the full p.d. and a failed bulb only breaks its own branch, as with the Nairobi tubes. (d) Kettle: $I = P/V = 3000 / 240 = 12.5$ A. Heater: $2000 / 240 = 8.33$ A. Appliances are in parallel, so their currents add: $12.5 + 8.33 = 20.8$ A, which is more than the 13 A fuse rating. The thin fuse wire heats and melts, breaking the circuit before the cable overheats. (e) The classroom should be wired in parallel with a fuse or circuit breaker on the live wire to cut off excess current from an overload or short circuit, and with earthing to give fault current a safe path and protect people from shock.
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Part 5: Final explanation and model revision

Write a complete final explanation that answers the driving question: Why do the other tubes stay lit when one blows out, and why do bulbs connected in series glow dimmer than identical bulbs connected in parallel to the same battery? Use evidence from at least three of our lessons (for example the current and p.d. measurements, Ohm's law, and resistance in series and in parallel). Then state one idea in your Lesson 1 model that you now want to change, and say how you would change it. Finally, say how this helps you answer the KICD questions: how do we determine the relationship between current, p.d. and resistance, and how is current	The other tubes stay lit because they are connected in parallel: each tube is on its own branch directly across the 240 V supply. When one branch is broken, the others still have the same p.d. and the same current (0.15 A each; the total falls from 0.60 A to 0.45 A). Series bulbs are dim because the three identical bulbs share the 9 V of the pack (3.0 V each) in one loop of total resistance $30 + 30 + 30 = 90 \Omega$. By $V = IR$ the current is only $9/90 = 0.10$ A, so each bulb transfers $P = IV = 3.0 \times 0.10 = 0.30$ W. In parallel each bulb has the full 9 V across 30Ω, so the current is 0.30 A and the power 2.7 W, which is 9 times larger, and the effective resistance is 10Ω with 0.90 A from the battery. In a series loop a failed bulb opens the only path so everything goes out, as in the old Christmas-light string. Evidence: ammeter readings of 0.10 A and 0.30 A (Lesson 3), voltmeter readings of 3.0 V and 9.0 V (Lesson 4), the V-I graph for the 30Ω resistor (Lesson 5), and the resistance rules (Lessons 7 and 8). Model change: in my first model I thought the battery pushed the same amount into every bulb, or that bulbs used up current; now I show the same current in the series loop, the 9 V shared as 3.0 V per bulb, and separate branches each with the full p.d. in parallel. KICD question 1: we found the relationship by measuring V and I for the 30Ω resistor; V/I stayed 30Ω, so $V = IR$, and current rises with p.d. and falls with resistance. KICD question 2: current electricity is applied in home and school wiring, with lights in parallel, fuses or breakers on the live wire, earthing, and choosing efficient LEDs to cut the cost of energy in kWh.
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electricity applied in day-to-day life?	
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of evidence and data	Uses the exact values from the demos and classroom (90 Ω , 10 Ω , 0.10 A, 0.30 A, 3.0 V, 9.0 V, 0.30 W, 2.7 W, 0.15 A, 0.60 A, 0.45 A) accurately and links them to claims.	Uses most of the correct values but some are missing or not linked clearly to the claim.	Uses few or incorrect values, or gives claims without evidence.
Scientific reasoning about series and parallel circuits	Clearly explains that series bulbs share the p.d. and a single current through 90 Ω , while parallel branches each get the full p.d. and independent current, and links this to why one blown tube does not affect the others.	Explains the series and parallel difference correctly but with limited reasoning about p.d., current or resistance.	Describes what happens without a correct reason, or shows misconceptions such as bulbs using up current.
Calculations and units	All calculations ($V = IR$, resistance in series and parallel, $P = IV$, $E = Pt$, cost) are set out step by step with correct units and answers.	Calculations mostly correct with minor errors in units or steps.	Calculations incomplete or incorrect, or missing units.
Application and safety	Correctly explains the fuse blowing at 20.8 A against 13 A, the roles of the fuse or breaker and earthing, and the cost saving of an LED, in Kenyan contexts.	Explains most safety and cost ideas correctly with some gaps.	Safety and cost ideas are unclear or missing.
Model revision and communication	Compares the first model with the final model, names a specific change, links to both KICD questions, and uses scientific terms (in series, in parallel, p.d., e.m.f.) correctly.	Describes a change in the model and uses most terms correctly.	Does not describe a model change or uses terms incorrectly.

ATTRIBUTION AND LICENSING

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Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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